Ultrafast Quantum Repeaters for Long Distance Quantum Communication

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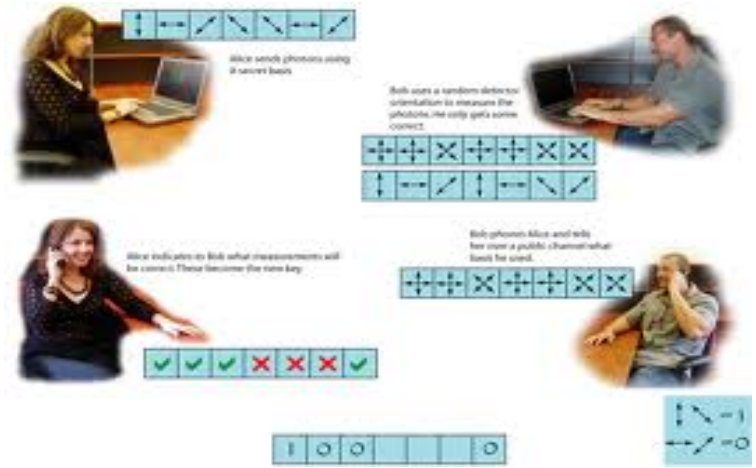
Sreraman Muralidharan (Yale), Linshu Li (Yale)
Norbert Lütkenhaus (Waterloo), Jungsang Kim (Duke), Misha Lukin (Harvard)

Qcrypt 2015
(2015.10.1)

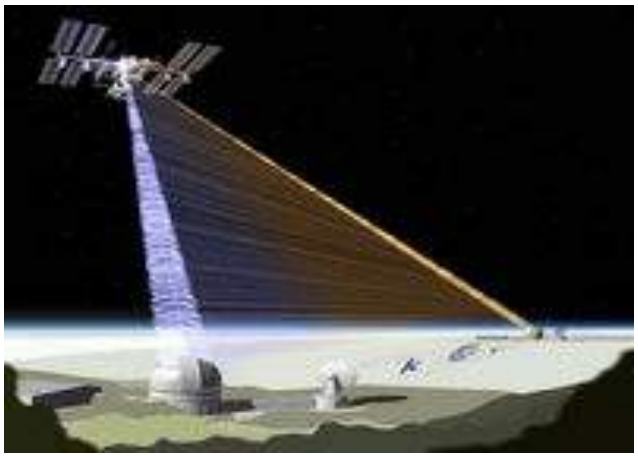
Supported by: DARPA-Quiness, ARL-CDQI, ARO, AFOSR, Sloan, Packard.

Quantum Communication

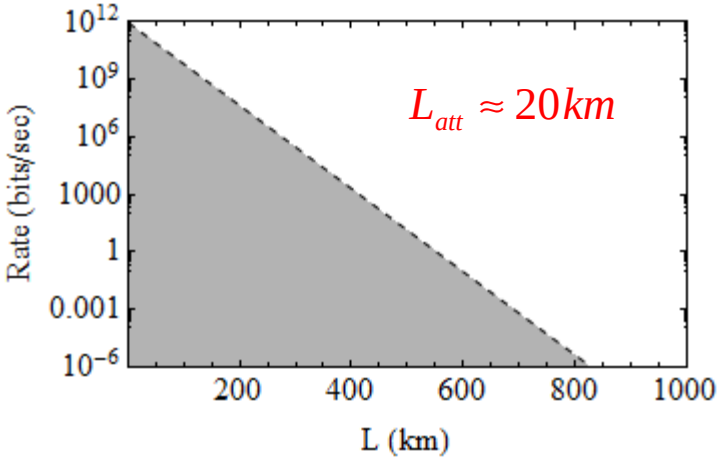
Quantum Key Distribution (e.g., BB84, Ekert91)



Solution 1: Satellite based QKD

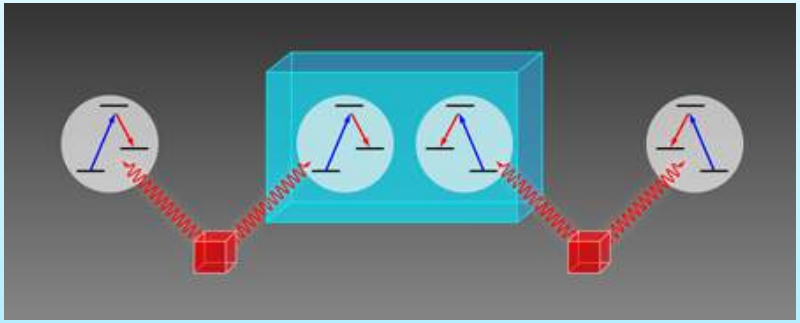


Major Challenge – Fiber Attenuation



M. Takeoka, S. Guha and M. M. Wilde, Nat. Commun. **5** (2014).

Solution 2: Quantum Repeaters

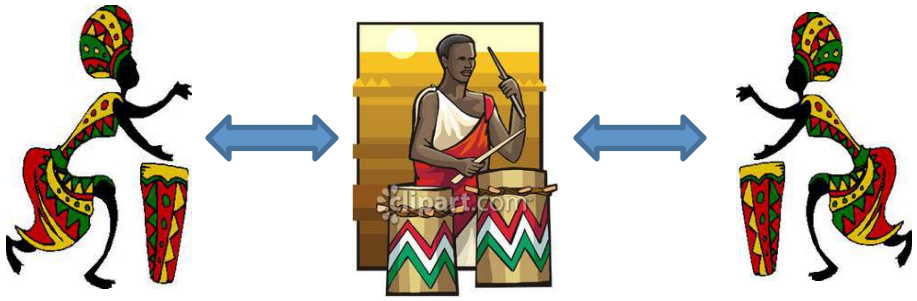


Classical Repeaters

Repeaters based on Smoke Signal (Great Wall, 900BC)



Repeaters based on Sound (Africa Drum Communication)



Repeaters based on Optical Signal (Undersea Cables)

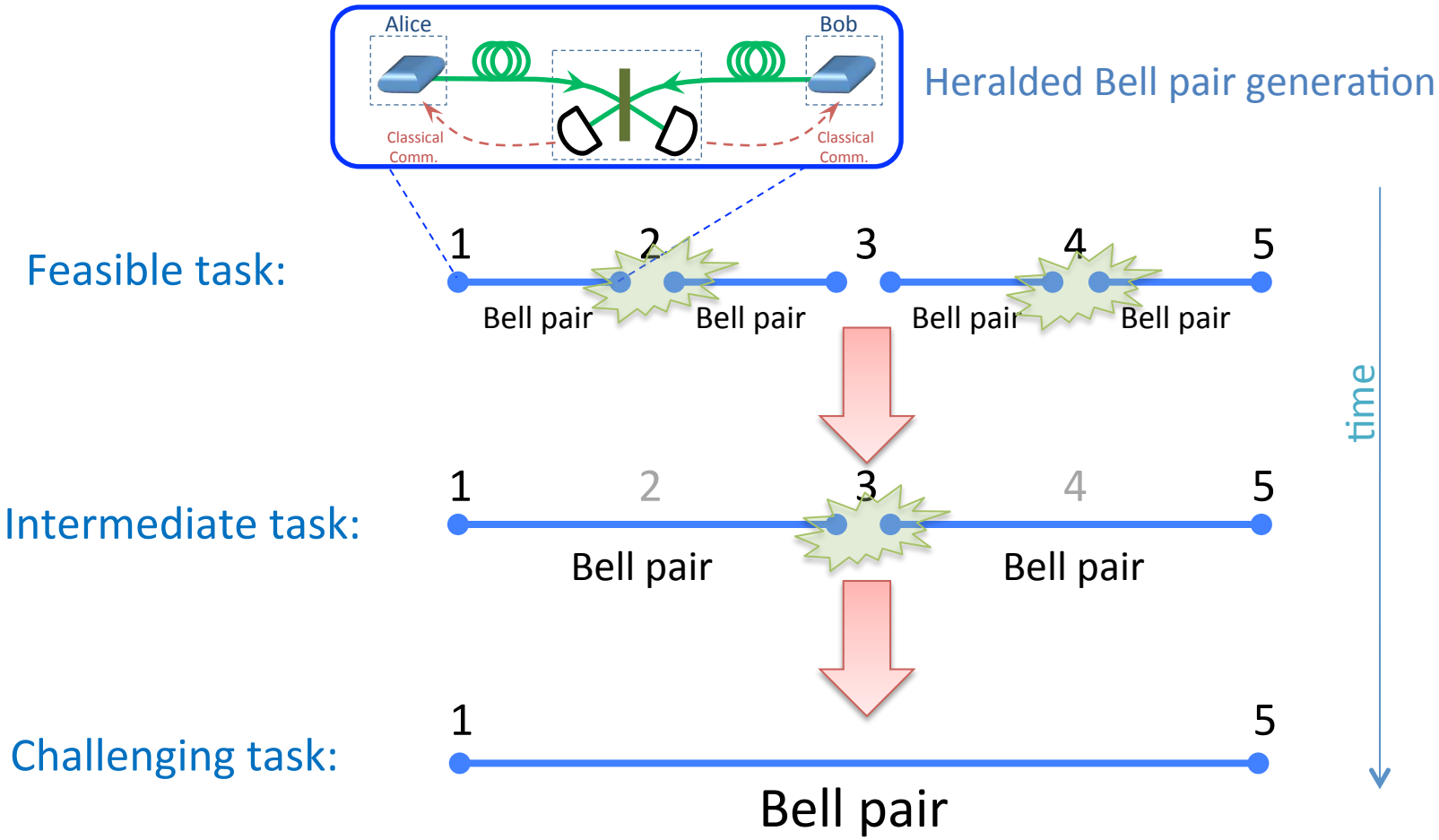


Our Quantum World

- **Challenges**: Quantum No-Cloning Theorem
  - Unknown quantum states cannot be perfectly cloned
- **Opportunities**: Quantum Entanglement
  - Quantum state teleportation
  - Entanglement swapping
  - Non-local coupling gate
- **Imperfections**:
  - *Loss Errors* (Fiber loss L_{att}≈20km, coupling & detector inefficiency)
  - *Operation Errors* (Channel decoherence, memory errors, local gate/ measurement errors)

Key Ideas for Quantum Repeaters

- To overcome ***loss errors***

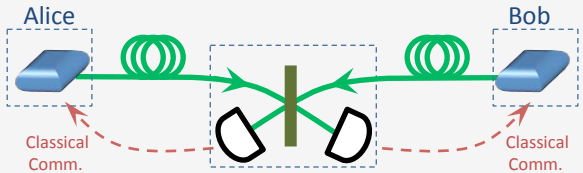
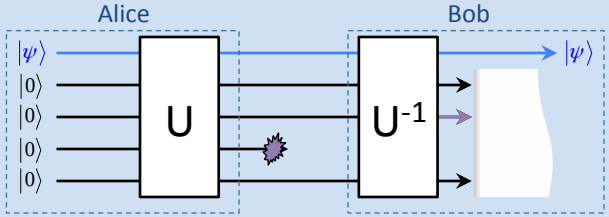
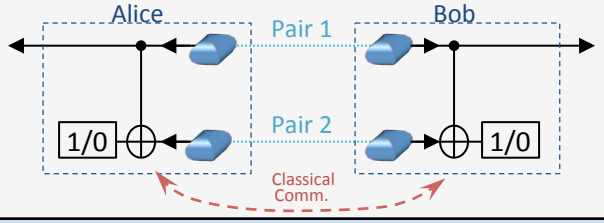
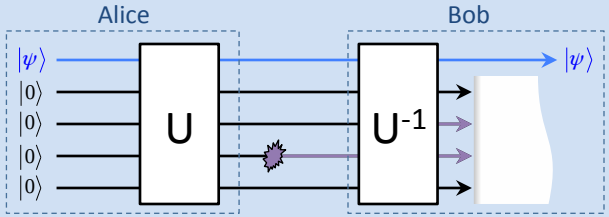


- For ***operation error ε***, final infidelity $\sim \varepsilon N \sim \varepsilon \frac{L_{tot}}{L_{att}} < 1$

$$L_{tot} < L_{att} / \varepsilon$$

THREE GENERATIONS OF QRS

How to Overcome Both Loss & Operation Errors?

	Approaches	Example	Requirement
Loss Errors	**Heralded Generation (*)**		Prob. & Heralded, ***Two-Way*** Comm.
	Quantum Error Correction		Deterministic, ***One-Way*** Comm. Suppress $\varepsilon \rightarrow \varepsilon^{2t+1}$
Operation Errors	**Heralded Purification (**)**		Prob. & Heralded, ***Two-Way*** Comm.
	Quantum Error Correction		Deterministic, ***One-Way*** Comm. Suppress $\varepsilon \rightarrow \varepsilon^{t+1}$

(*) Experiments with ions, atoms, NVs, QDs, ensembles, … (**) Experiments with photons, ions, …

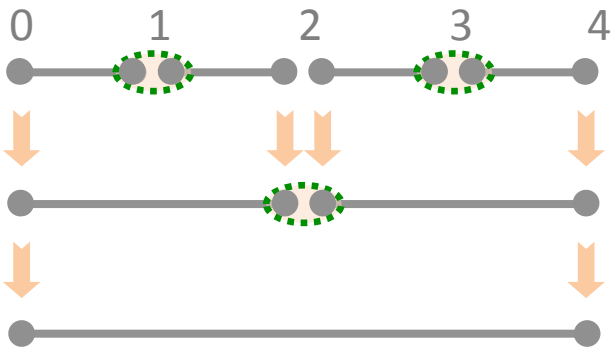
Three Generations of QRs

	Approaches	1st Generation	2nd Generation	3rd Generation
Loss Errors	Heralded Generation [*Two-Way* Comm.]			
	Quantum Error Correction [*One-Way* Comm.]			
Operation Errors	Heralded Purification [*Two-Way* Comm.]			
	Quantum Error Correction [*One-Way* Comm]			

1St Generation QRs

Key Idea: Nested self-similar architecture of heralded ent. generation & purification.

- Procedure
  - ***Heralded*** Entanglement Generation (loss errors)
  - Connection
  - ***Heralded*** Entanglement Purification (opration errors)
- Time Scaling of Rate
  - ***1/Poly(L_{tot})***
- Challenges
  - ***Slow*** for long distances, even after optimization
    [L.J., Taylor, Khaneja, Lukin. PNAS 104, 17291 (2007)]
  - ***Significant*** memory errors *[Hartmann, Kraus, Dür, Briegel. PRA 75, 032310 (2007)]*
- Boost with multiplexing for intermediate distance (~1000km)
  - Spatial, temporal, frequency multiplexing
    *[Sangouard, Simon, de Riedmatten, Gisin, RMP **83**, 33 (2011)]* & ***Hugues de Riedmaten's Talk***.
- Limitation
  - ***Heralded*** Entanglement Purification
    -> ***Two-way*** communication,
    -> slows for long distances (>1000km)



[Briegel, Dür, Cirac, Zoller. PRL 81, 5932 (1998)]
[Duan, Lukin, Cirac, Zoller, Nature 414, 413 (2001)

Three Generations of QRs

	Approaches	1st Generation	2nd Generation	3rd Generation
Loss Errors	Heralded Generation [*Two-Way* Comm.]			
	Quantum Error Correction [*One-Way* Comm.]			
Operation Errors	Heralded Purification [*Two-Way* Comm.]			
	Quantum Error Correction [*One-Way* Comm]			

2nd Generation QRs

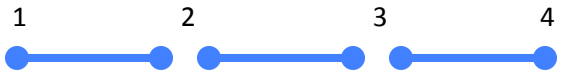
Key Idea: Create almost perfect Bell pairs at the encoded level

Without Encoding

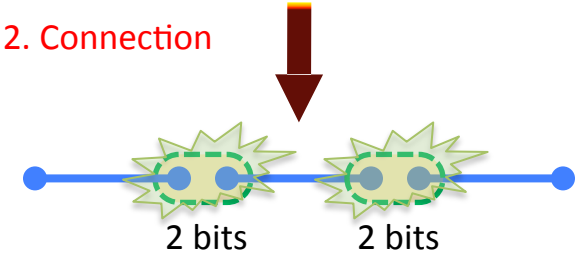
(Operation Error: ε)

time

1. Generation



2. Connection



3. Entanglement swapping



Final Infidelity:

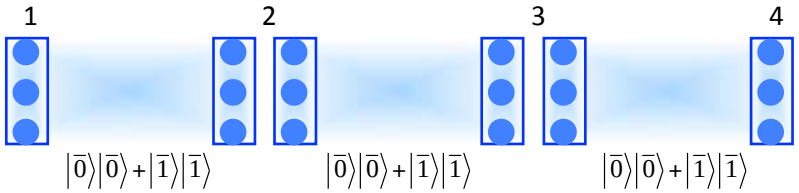
$$1-(1-\varepsilon)^N \approx N\varepsilon$$

$$L_{tot} \sim L_{att} / \varepsilon$$

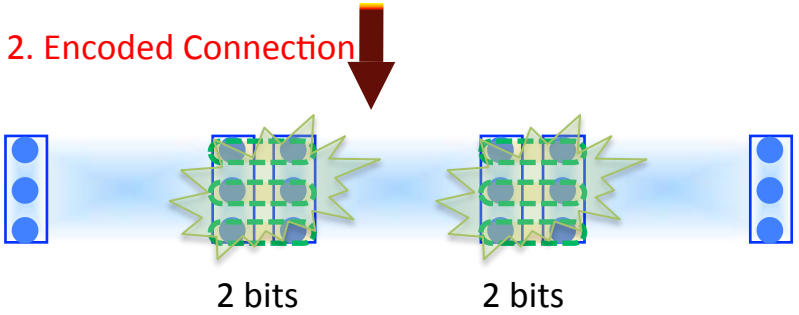
With Quantum Encoding [[n,1,2t+1]]

(Suppressed Opr. Errors: ε-> ε^{t+1})

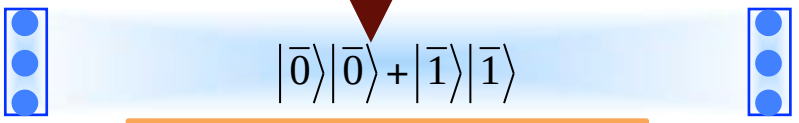
1. Encoded Generation



2. Encoded Connection



3. Encoded Entanglement swapping



Final Infidelity:

$$1-\left(1-\varepsilon^{t+1}\right)^N \approx N\varepsilon^{t+1}$$

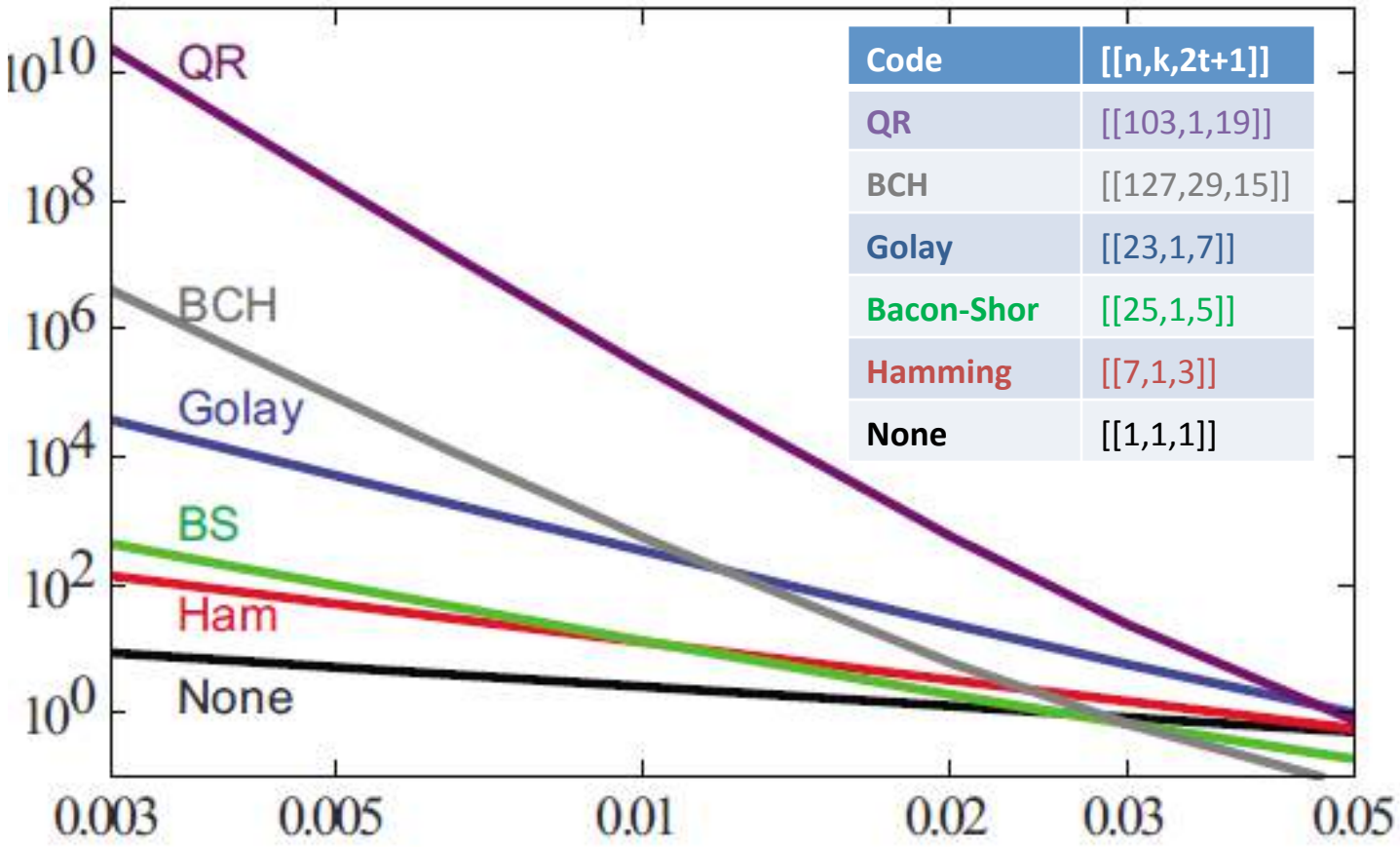
$$L_{tot} \sim L_{att} / \varepsilon^{t+1}$$

[L.J., Taylor, Nemoto, Munro, Van Meter, Lukin, PRA 79, 032325 (2009)]

2nd Generation QRs

Maximum number of repeater stations

$N = L_{tot} / L_0$



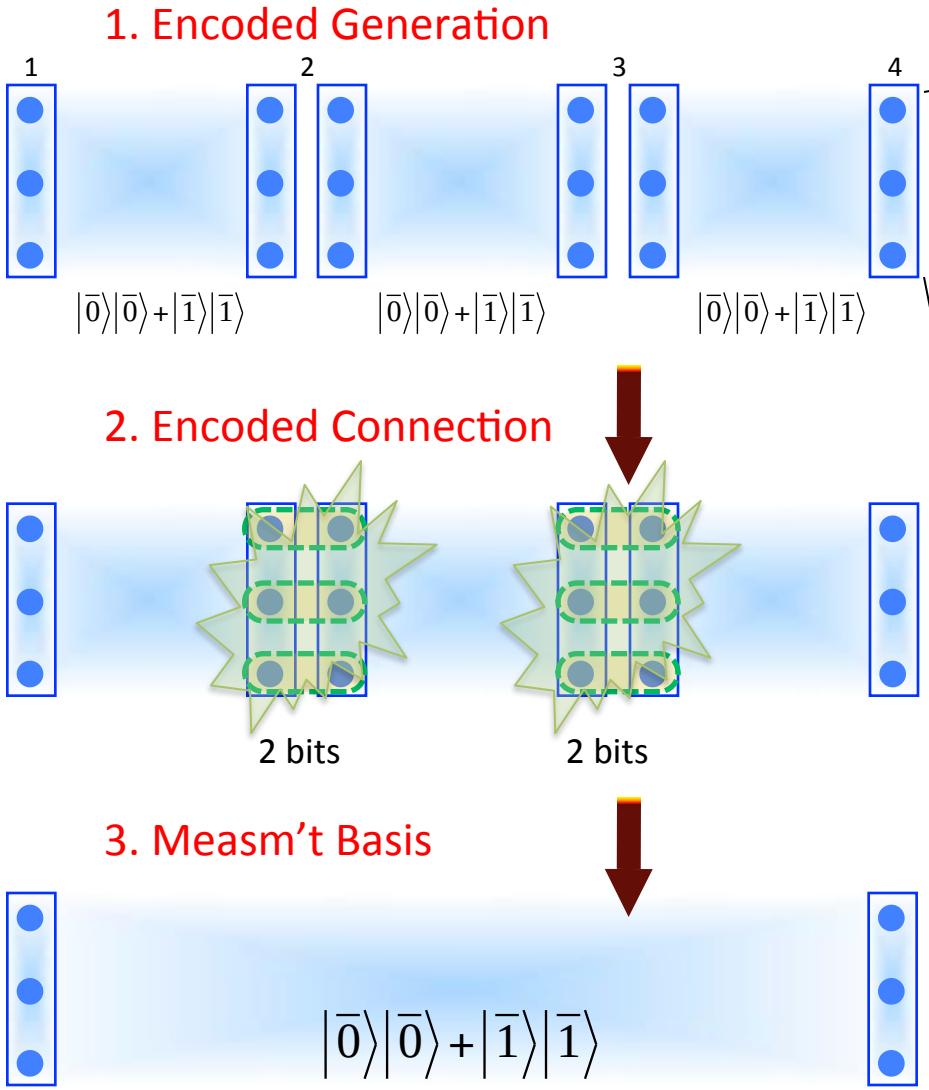
Code	[[n,k,2t+1]]
QR	[[103,1,19]]
BCH	[[127,29,15]]
Golay	[[23,1,7]]
Bacon-Shor	[[25,1,5]]
Hamming	[[7,1,3]]
None	[[1,1,1]]

Effective error probability $q \sim 5\varepsilon$

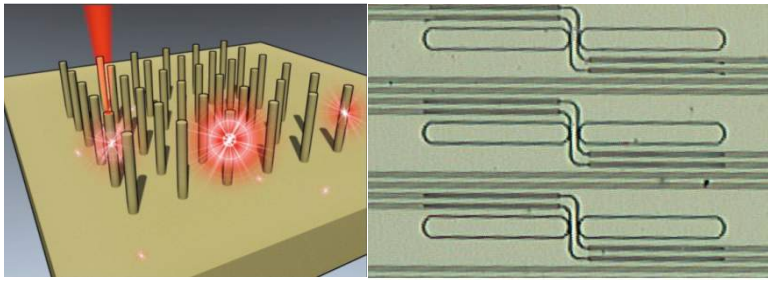
[L.J., Taylor, Nemoto, Munro, Van Meter, Lukin, PRA 79, 032325 (2009)]

2nd Generation QRs

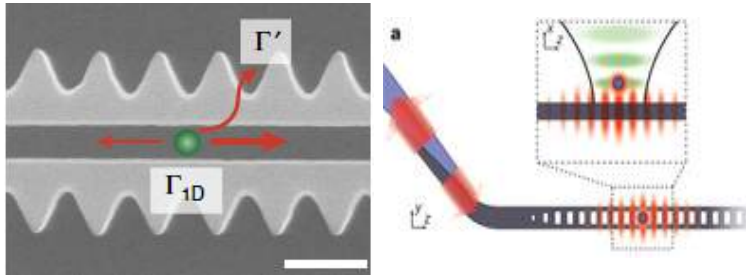
-- *Potential Implementations with NV/Ion/Atom/QD*



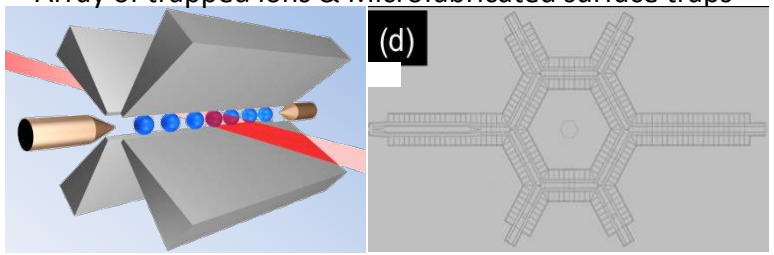
Array of NV centers & Integrated on-chip waveguide



Array of trapped atoms & Nano-photonics



Array of trapped ions & Microfabricated surface traps



[L.J., Taylor, Nemoto, Munro, Van Meter, Lukin, PRA 79, 032325 (2009)]

2nd Generation QRs

- Key Idea:
  - Create almost perfect Bell pairs at the encoded level (CSS code)
    [L.J., Taylor, Nemoto, Munro, Van Meter, Lukin, PRA 79, 032325 (2009)]
- Procedure
  - ***Heralded*** Entanglement Generation/Purification [loss errors]
  - ***Deterministic*** Entanglement Connection & Error Correction [operation errors]
- Time Scaling of Rate
  - *1/[τ_0*Poly(Log(L_{tot}))]*, with pair generation time τ_0

$$\tau_0 \propto \frac{l_0}{c}\frac{\exp\left(l_0 / l_{att}\right)}{\eta^2} \approx \frac{0.1ms}{\eta^2} \quad \text{for repeater spacing } l_0 = 10km.$$

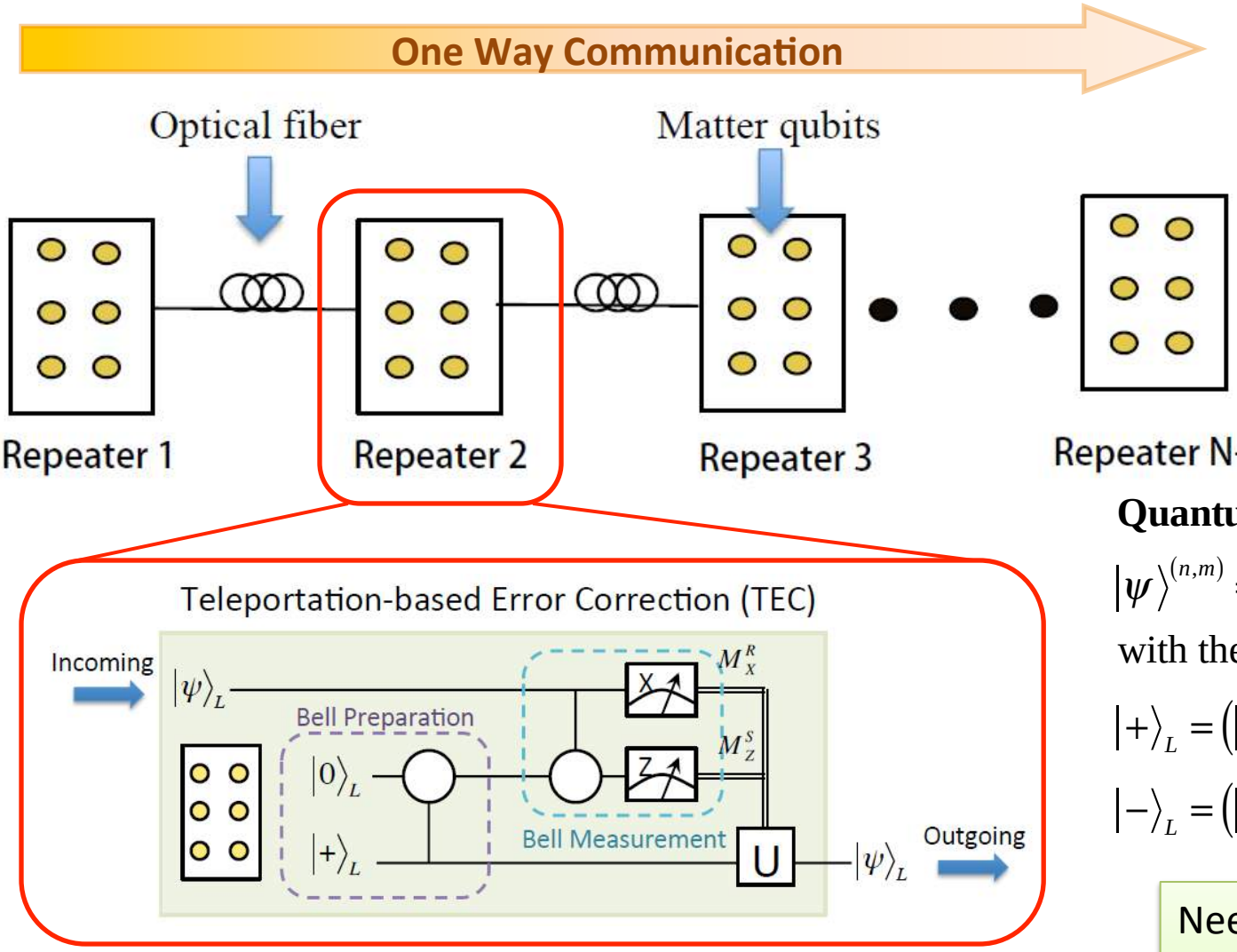
- Advantages
  - **Fast** for **long** distances (>Kbits/sec)
  - **Suppressed** memory errors *[Munro, et al., Nature Photonics 4, 792 (2010)]*
- Limitation
  - ***Heralded*** Entanglement Generation still requires ***two-way*** communication between neighboring stations to suppress loss errors, which limit τ_0.
  - Can we design efficient QR protocols to overcome loss errors?

Three Generations of QRs

	Approaches	1st Generation	2nd Generation	3rd Generation
Loss Errors	Heralded Generation [*Two-Way* Comm.]			
	Quantum Error Correction [*One-Way* Comm.]			
Operation Errors	Heralded Purification [*Two-Way* Comm.]			
	Quantum Error Correction [*One-Way* Comm]			

3rd Generation QRs

-- Architecture based on Quantum Error Correction



Quantum Parity Code :

$$|\psi\rangle^{(n,m)} = \alpha|+\rangle_L + \beta|-\rangle_L$$

with the logiacl basis:

$$|+\rangle_L = \left(|00\ldots0\rangle_{1\ldots m} + |11\ldots1\rangle_{1\ldots m}\right)^{\otimes n}$$

$$|-\rangle_L = \left(|00\ldots0\rangle_{1\ldots m} - |11\ldots1\rangle_{1\ldots m}\right)^{\otimes n}$$

Need correct outcomes:
$\tilde{M}_X$ & $\tilde{M}_Z$
to complete state transfer.

Munro, Stephens, Devitt, Harrison, Nemoto, Nature Photonics 6, 777 (2012)

Muralidharan, Kim, Lütkenhaus, Lukin, L.J., PRL 112, 250501 (2014)

Measurement Outcomes

Distribution of Outcomes

$\tilde{M}_X \setminus \tilde{M}_Z$	+1	-1	0
+1	$p_{1,1}$	$p_{1,-1}$	$p_{1,0}$
-1	$p_{-1,1}$	$P_{-1,-1}$	$P_{-1,0}$
0	$p_{0,1}$	$p_{0,-1}$	$P_{0,0}$

Heralded failure

Heralded Success (for *N* stations):

$$P_{succ} = \left(p_{1,1} + p_{1,-1} + p_{-1,1} + p_{-1,-1}\right)^N$$

QBER for X/Z measurement

$$Q_{X/Z} = \frac{1}{2}\left[1 - \left(\frac{p_{1,1} \pm p_{1,-1} \mp p_{-1,1} - p_{-1,-1}}{p_{1,1} + p_{1,-1} + p_{-1,1} + p_{-1,-1}}\right)^N\right]$$

Secure Key Generation Rate:

$$R = \frac{1}{\tau_{opr}} \max\left[P_{succ}\left(1 - 2h\left(\frac{Q_X + Q_Z}{2}\right)\right), 0\right]$$

with local gate operation time τ_{opr}, and binary entropy function $h(Q)$.

Muralidharan, Kim, Lütkenhaus, Lukin, L.J., PRL 112, 250501 (2014)
Scarani, Pasquinucci, Cerf, Dusek, Lütkenhaus, Peev, RMP. 81, 1301 (2009).

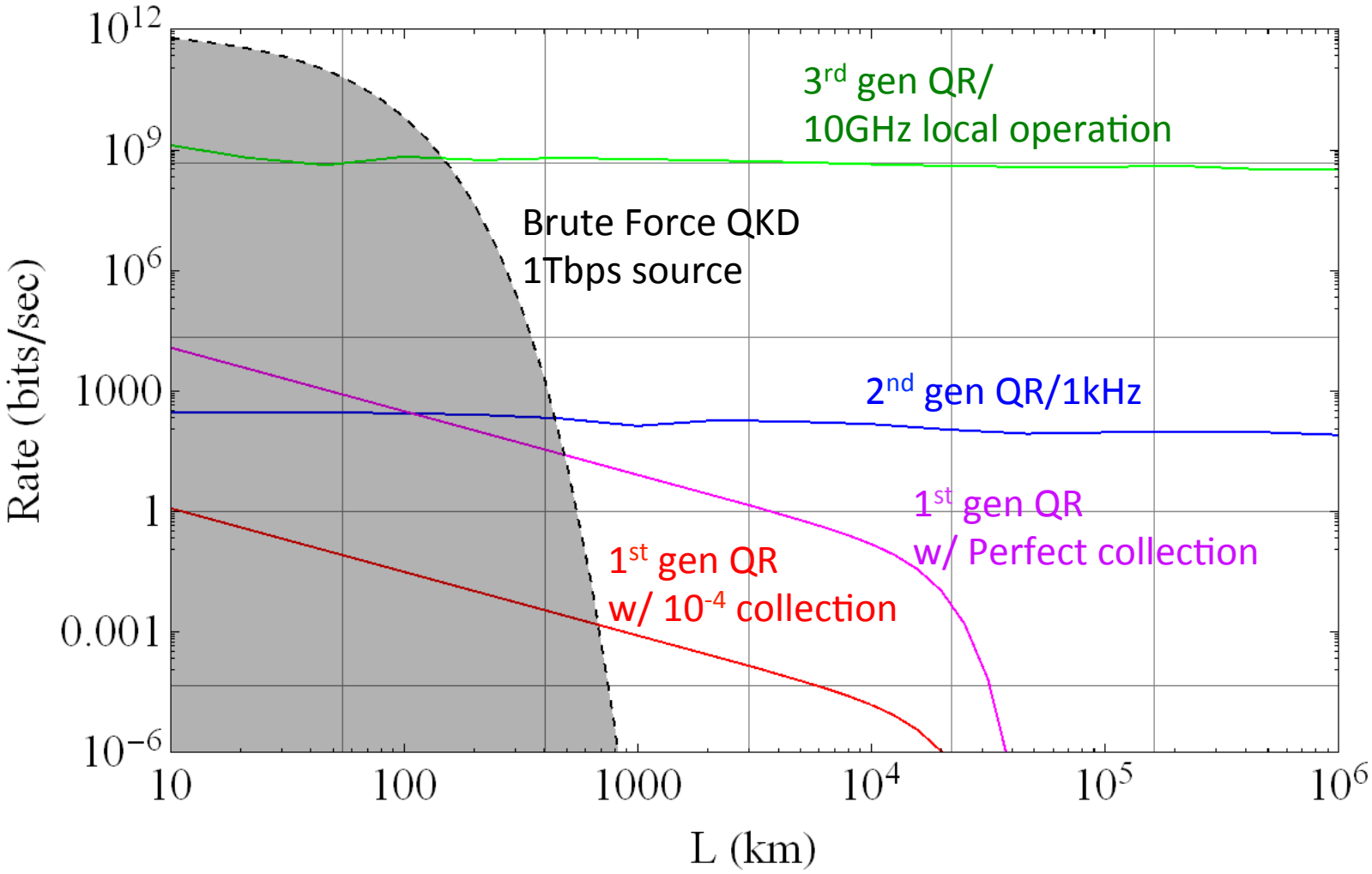
3rd Generation QRs

- Key Idea:
  - Use Quantum Error Correction for **BOTH** loss and operation errors
    [Fowler, Wang, Hill Ladd, Van Meter, Hollenberg, PRL 104, 180503 (2010)]
    [Munro, Stephens, Devitt, Harrison, Nemoto, Nature Photonics 6, 777 (2012)]
    [Muralidharan, Kim, Lutkenhaus, Lukin, L.J., PRL 112, 250501 (2014*)]*
- Procedure
  - Quantum Encoding & Send/Propagate Quantum States over Optical Channel
  - Apply Quantum Error Correction (loss errors & operation errors)
- Time Scaling of Rate
  - $1/[\tau_{opr} * Poly(Log(L_{tot}))]$, with local gate operational time τ_{opr}
- Advantages
  - **Ultra Fast** for long distances (>Mbits/sec)
  - **Suppressed** memory errors *[Munro, et al., Nature Photonics 6, 777 (2012)]*
  - Possibility of **all-photonic QRs** *[Azuma, Tamaki, Lo,, Nat Commun 6, 6768 (2015)]*
- Challenges
  - Local operation takes time, which limits *R*.
  - Large encoding block (tens or hundreds of qubits per node)
  - No more than 50% loss (no-cloning)

Three Generations of QRs

	Approaches	1st Generation	2nd Generation	3rd Generation
Loss Errors	Heralded Generation [*Two-Way* Comm.]			
	Quantum Error Correction [*One-Way* Comm.]			
Operation Errors	Heralded Purification [*Two-Way* Comm.]			
	Quantum Error Correction [*One-Way* Comm]			
Key Rate		$\ll \frac{c}{L_{tot}}$	$\sim \frac{c}{\eta^2 L_0}$	$\sim \frac{1}{\tau_{opr}}$

Rate Estimates



Cost Function & Cost Coefficient

Two types of resources to generate a secret key:

- **Time**: $1/R$
- **Qubits**: $n \times L_{tot} / L_0$

Cost = Time × Qubits: $C(L_{tot}) = \min_{n,L_0} \frac{1}{R} \frac{n L_{tot}}{L_0}$ [qubit*time/sbit]

Cost coefficient: $C'(L_{tot}) = C(L_{tot}) / L_{tot}$ [qubit*time/(sbit*km)]

Muralidharan, Kim, Lütkenhaus, Lukin, L.J., PRL 112, 250501 (2014)

Comparison Among Various QR Protocols

- Results of comparison among QR protocols:

Protocol Name	Properties & Refs
1st Gen	BDCZ scheme *[PRL 81, 5932 (1998)]*
2nd Gen (C)	Repeater with [[n,k,d]] CSS code *[PRA 79, 032325 (2009)]*
2nd Gen (NC)	Repeater with [[1,1,1]] CSS code *[PRA 79, 032325 (2009)]*
3rd Gen	Optimized quantum parity code *[PRL 112, 250501 (2014)]*

- The control parameters:

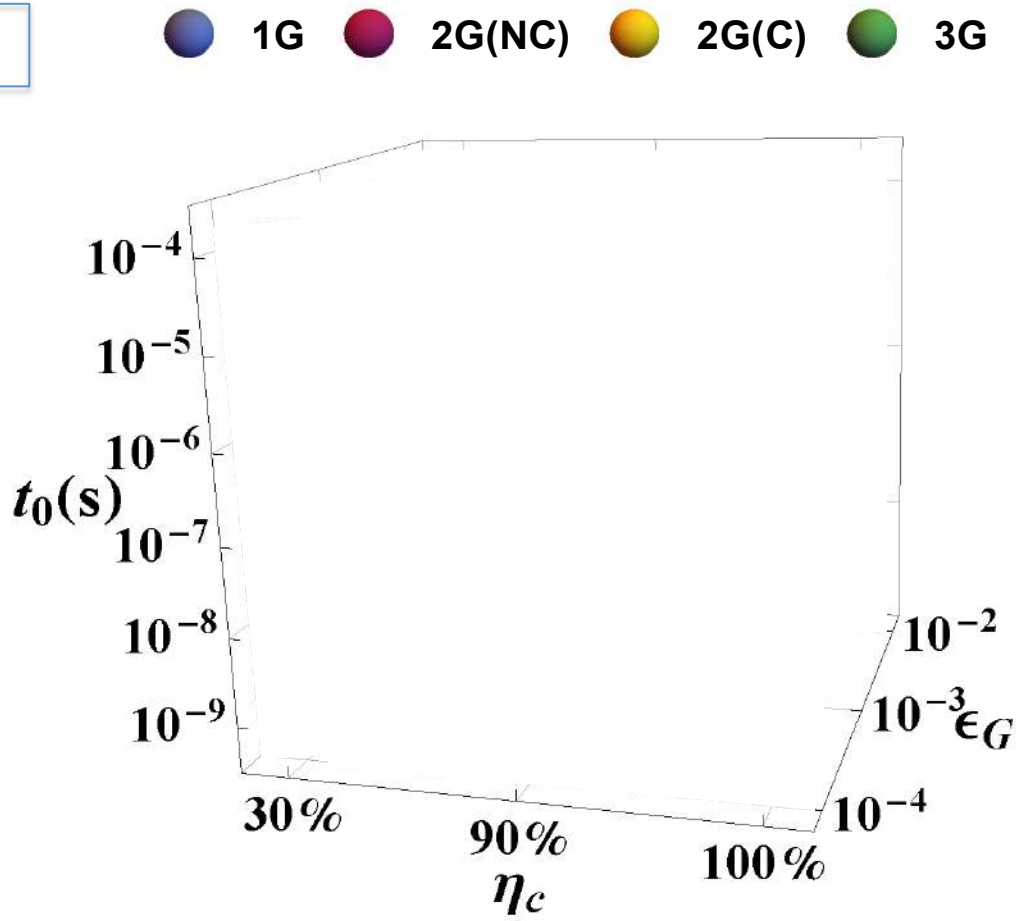
Parameter	Description	Typical values
L_{tot}	Total distance	{10^3km, 10^4km}
ε_G	Probability of gate error	(0.01% ~ 1%)
η_c	Coupling efficiency (between qubit and fiber)	{100%, 90%, 30%}
t_G	Time of local operations	(10^{-4} us ~ 10^3 us)

- Optimization criterion: Cost coefficient

$$C' \equiv \frac{qubit \times time}{sbit \times distance}$$

Optimum Generation of QRs

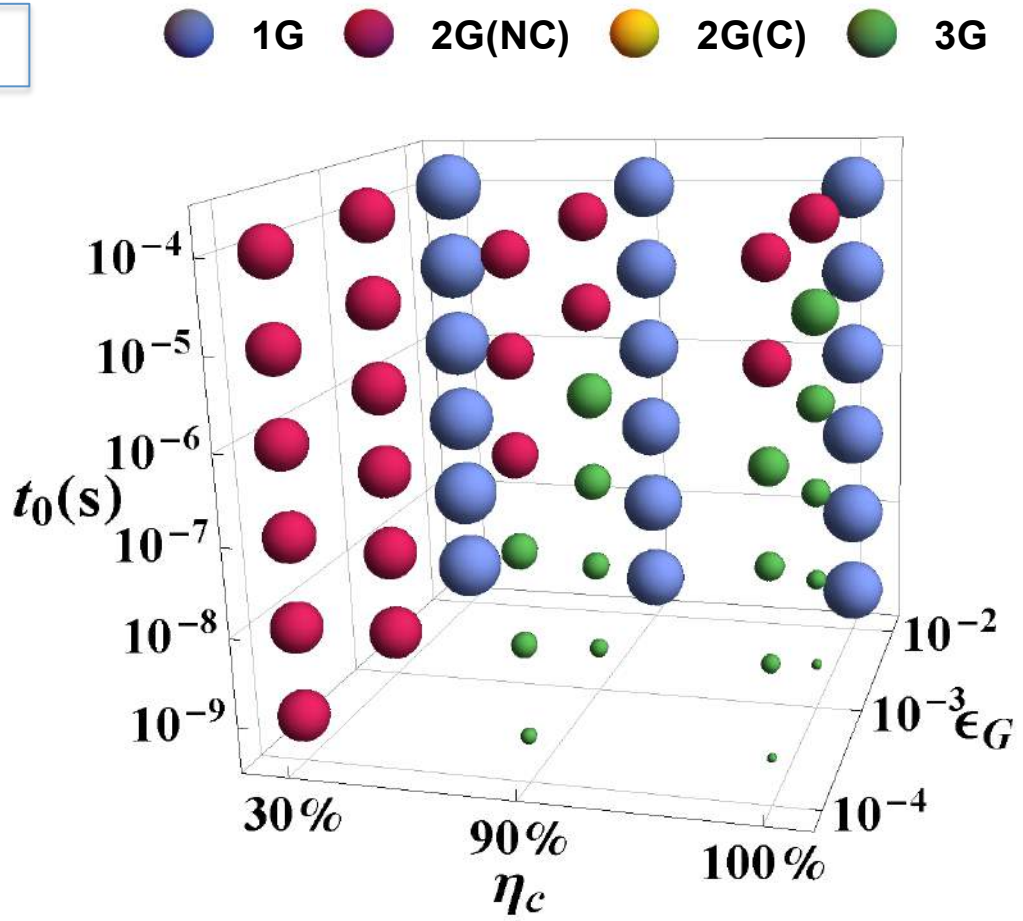
L_{tot} = 1000km



Muralidharan, Li, Kim, Lütkenhaus, Lukin, L. J., arXiv:1509.08135

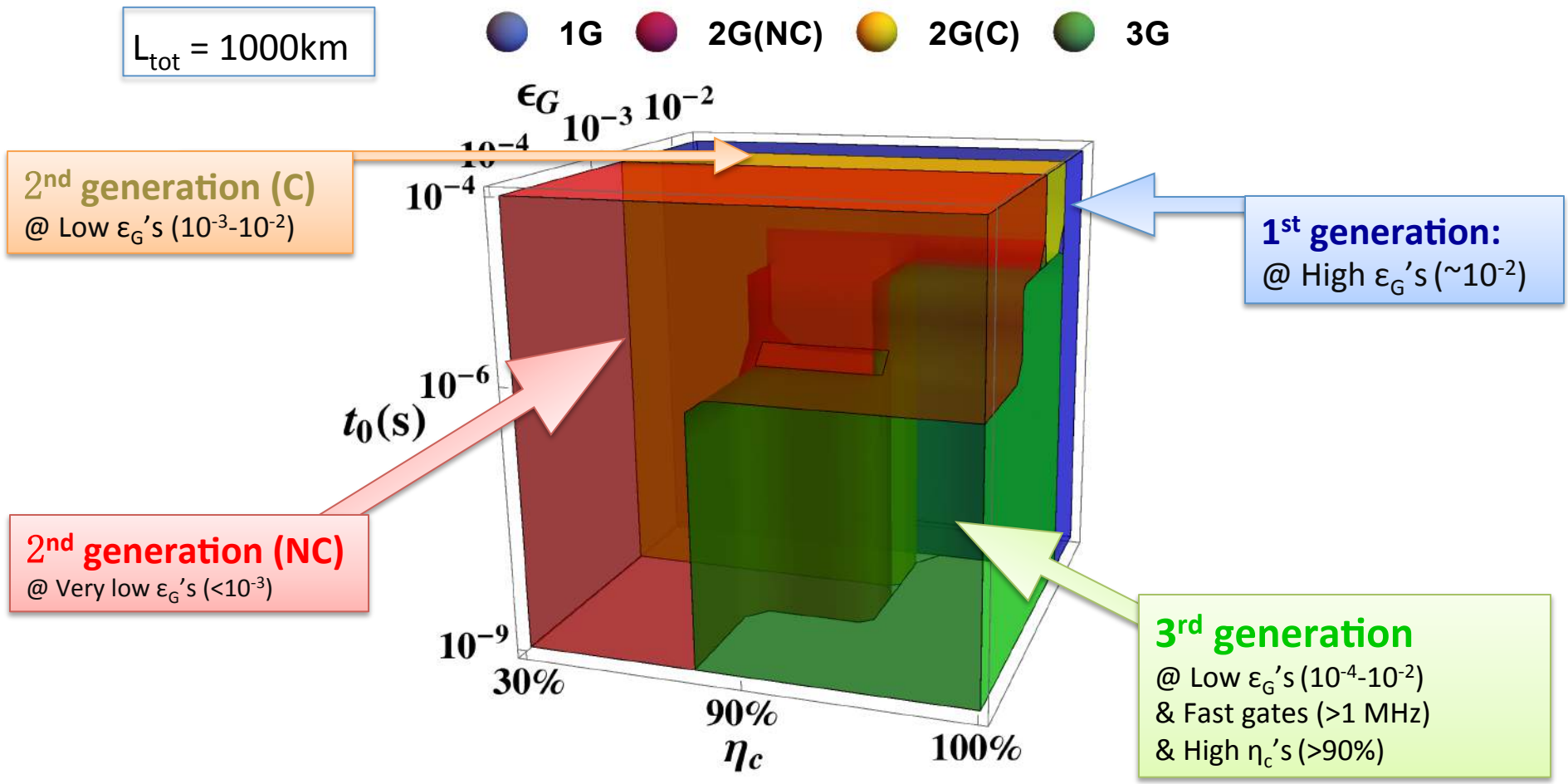
Optimum Generation of QRs

L_{tot} = 1000km



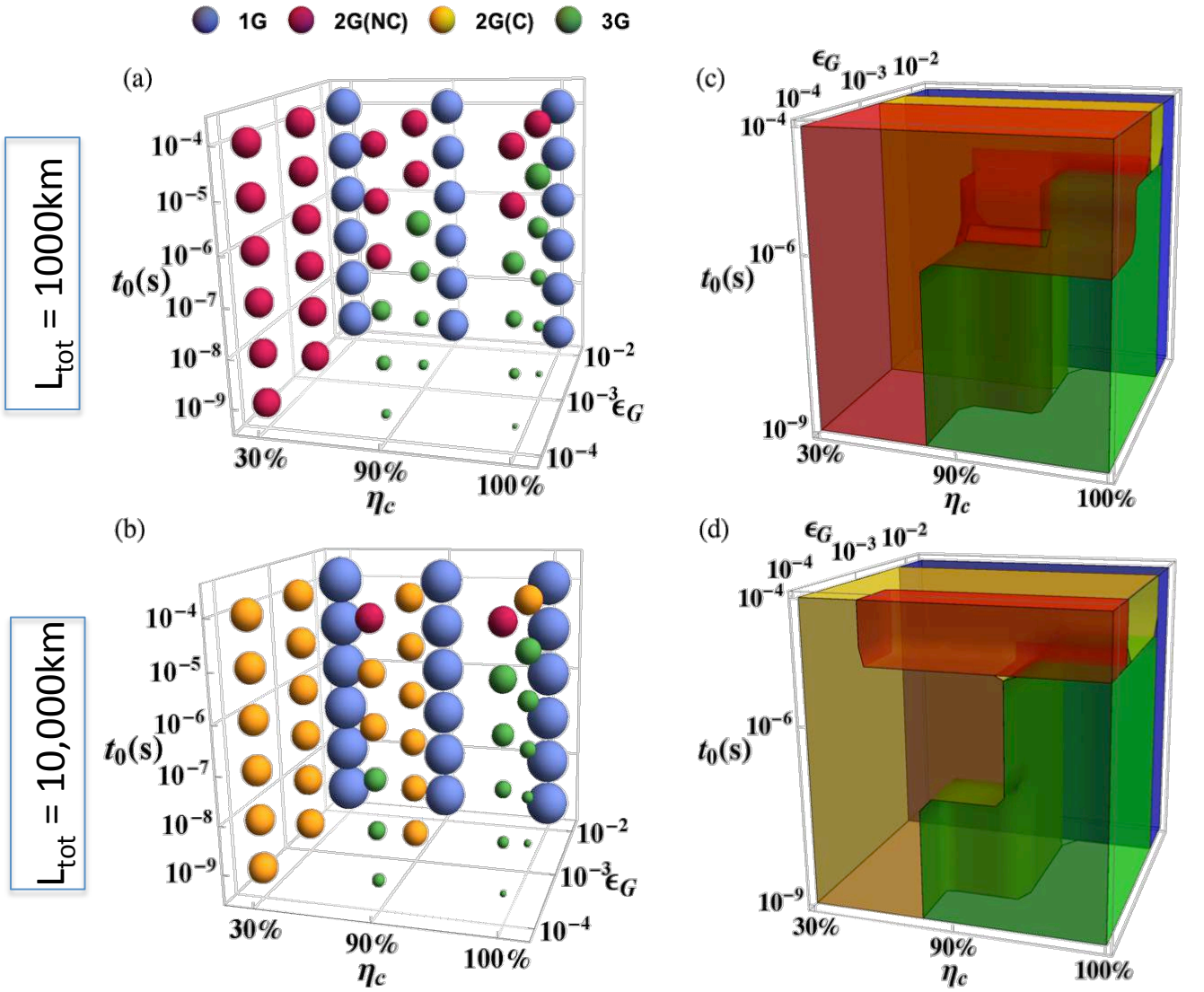
Muralidharan, Li, Kim, Lütkenhaus, Lukin, L. J., arXiv:1509.08135

Optimum Generation of QRs



Muralidharan, Li, Kim, Lütkenhaus, Lukin, L. J., arXiv:1509.08135

Optimum Generation of QRs



1st generation:

- High ε_G's (~10^{-2})
- Long L_{tot}'s (~10^4km)

2nd generation (NC)

- Low ε_G's (~10^{-4})
- Short L_{tot}'s (~10^3km)

2nd generation (C)

- Moderate ε_G's (10^{-4}-10^{-2})
- Long L_{tot}'s (~10^4km)

3rd generation

- Moderate ε_G's (10^{-4}-10^{-2})
- Long L_{tot}'s (~10^4km)
- Fast gates (>1 MHz)
- High η_c's (>90%)

Muralidharan, Li, Kim, Lütkenhaus, Lukin, L. J., arXiv:1509.08135

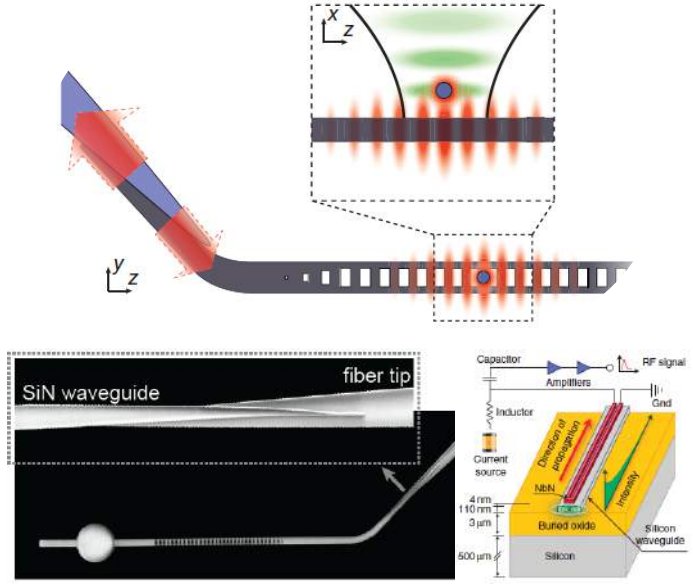
Parameters and Their Ranges

Quantities	Gate error rate*	Coupling efficiency	Local gate time
Symbols	ε_G	η_C	t_0
Ranges	0.01% ~ 1%	{30%, 90%, 100%}	1ns ~ 1 ms

* For simplicity, assume the measurement error is equal to gate error

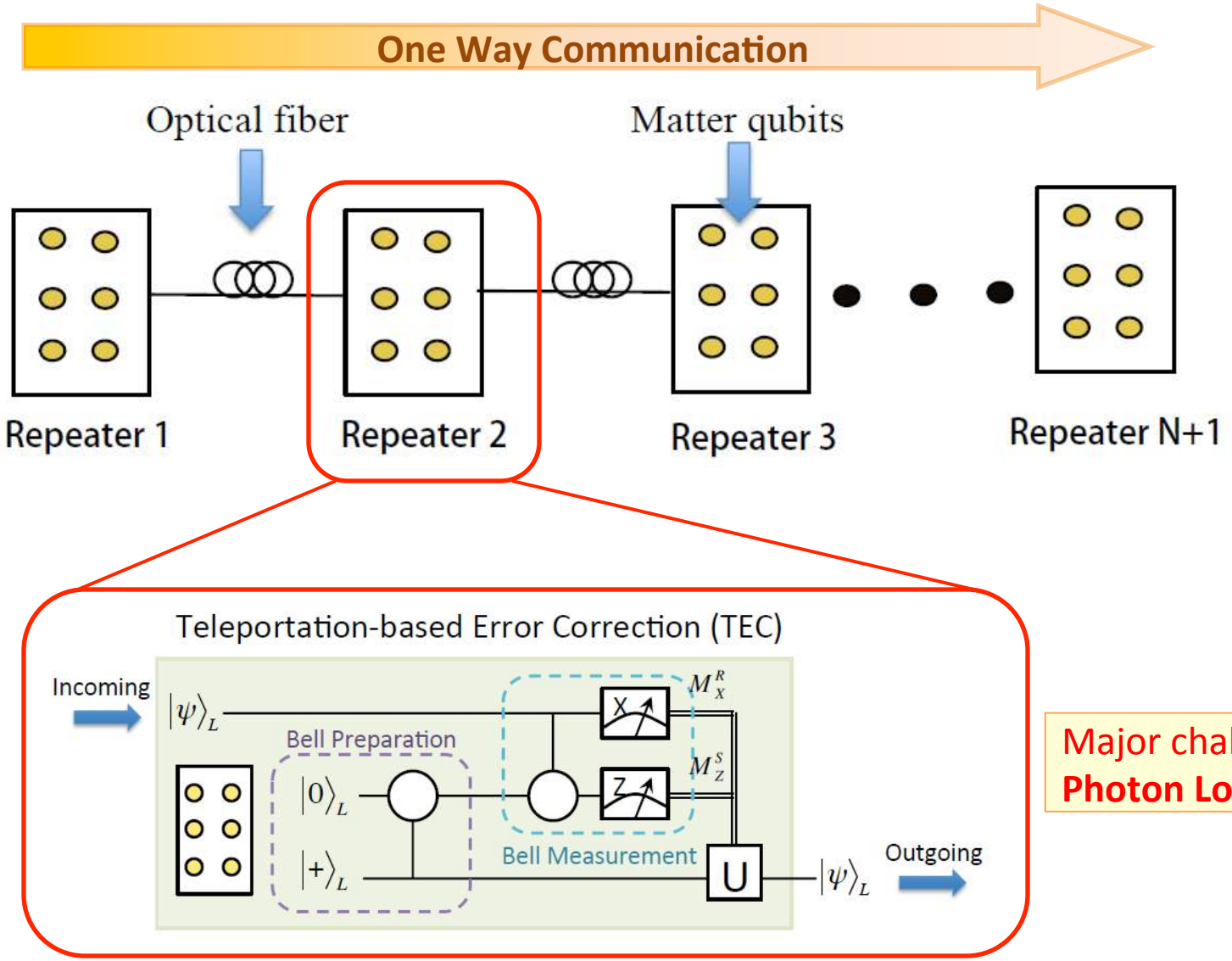
Recent Progress:

- ✓ Gate error $\varepsilon_G \sim 5*10^{-3}$ (trapped ion, SC qubits)
  Benhelm, et al, Nat Phys 4, 463 (2008). Barends, et al., Nature 508, 500 (2014)
- ✓ Single atom switch (nanophotonics, cavity)
  Tiecke, Thompson, et al., Nature 508, 241 (2014);
  Reiserer, et al., Nature 508, 237 (2014)
- ✓ Single photon detection(SSPD) efficiency $\eta_{det} > 90\%$
  W.H.P. Pernice et al., Nat. Comm. 3, 1325 (2012)
- ✓ Fiber-waveguide coupling efficiency $\eta_{FW} \geq 97\%$
  Tiecke, et al, Optica, 2, 70 (2015)



EFFICIENT QUANTUM CODES FOR QRS

Recall: Third Generation of QRs

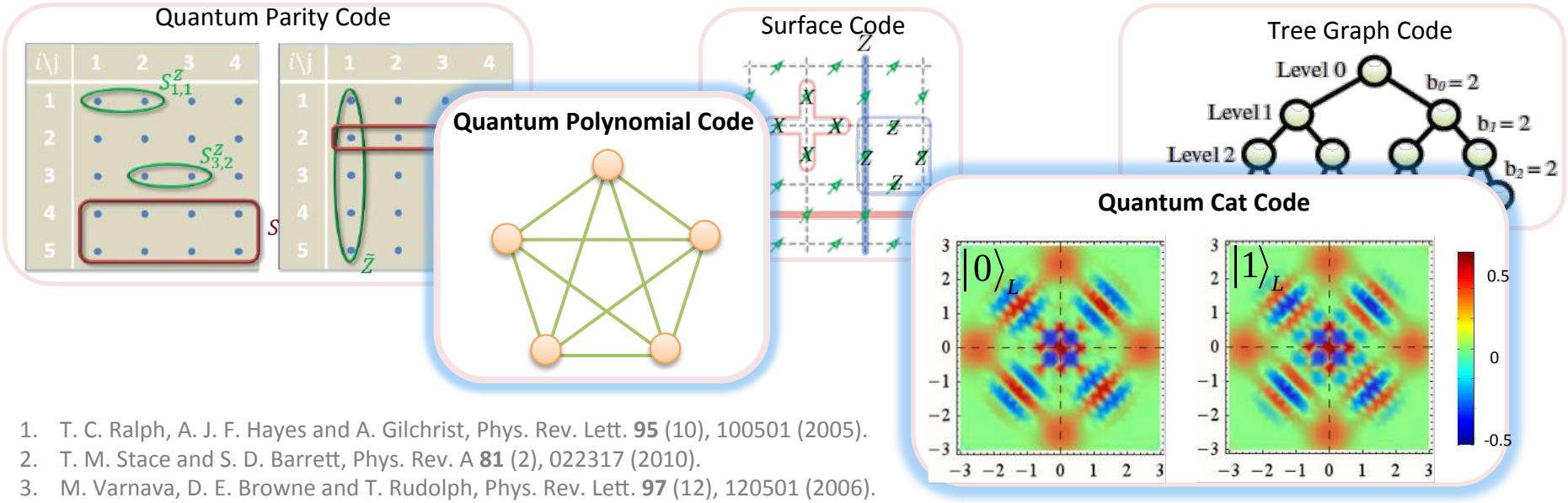


Major challenge:
Photon Loss Errors!

Munro, Stephens, Devitt, Harrison, Nemoto, Nature Photonics 6, 777 (2012)
*Muralidharan, Kim, Lütkenhaus, Lukin, L. J., PRL **112**, 250501 (2014)*

Various Coding Schemes

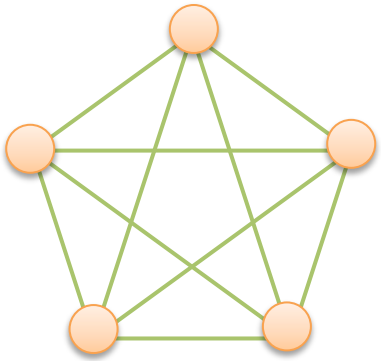
Physical Systems	Code Name	Loss Threshold
Qubits	Quantum parity code (QPC) [1]	~30%
	Surface code [2]	50%
	Tree graph code [3]	50%
Qudits	Quantum polynomial code (QPyC) [4]	50%
Continuous Variables	Cat code [5]	$O(n^{-1/2})$ for n>>1

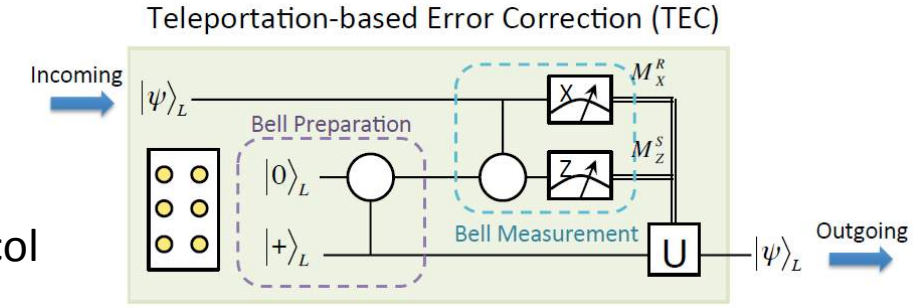


1. T. C. Ralph, A. J. F. Hayes and A. Gilchrist, Phys. Rev. Lett. **95** (10), 100501 (2005).
2. T. M. Stace and S. D. Barrett, Phys. Rev. A **81** (2), 022317 (2010).
3. M. Varnava, D. E. Browne and T. Rudolph, Phys. Rev. Lett. **97** (12), 120501 (2006).
4. R. Cleve, D. Gottesman and H.-K. Lo, Phys. Rev. Lett. **83** (3), 648-651 (1999).
5. M. Mirrahimi, Z. Leghtas, V. V. Albert, S. Touzard, R. Schoelkopf, L. J. and M. Devoret, New J. Phys. **16**, 045014 (2014).

Quantum Polynomial Code

- **Key Idea:** Using d-level quantum system (qudit) for encoding
- **Example:** 3-*qutrit* encoding $[[3,1,2]]_{d=3}$
- **Generalization: (**2k+1) *qudit* encoding $[[2k+1,1,k+1]]_d$
  - Restore $[[3,1,2]]_{d=3}$ for *k=1* & *d=3*.
  - Prime number *d>2k*
  - Decode from **any** *(k+1)* of the *(2k+1)* qudits
    [Cleve, Gottesman, Lo, PRL 83, 648 (1999)]
- **Properties:**
  - Correct loss up to $\frac{k}{2k+1}$, approaching non-cloning limit of 50% for large *k*.
  - CSS code, compatible with our TEC protocol
- **Implementation:**
  - E.g., Using single photon to encode d-level quantum system

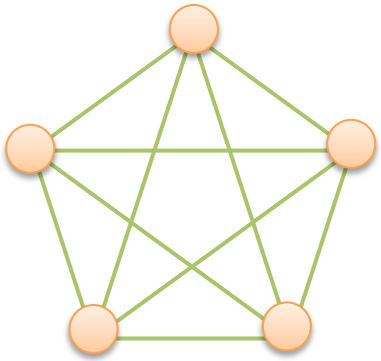




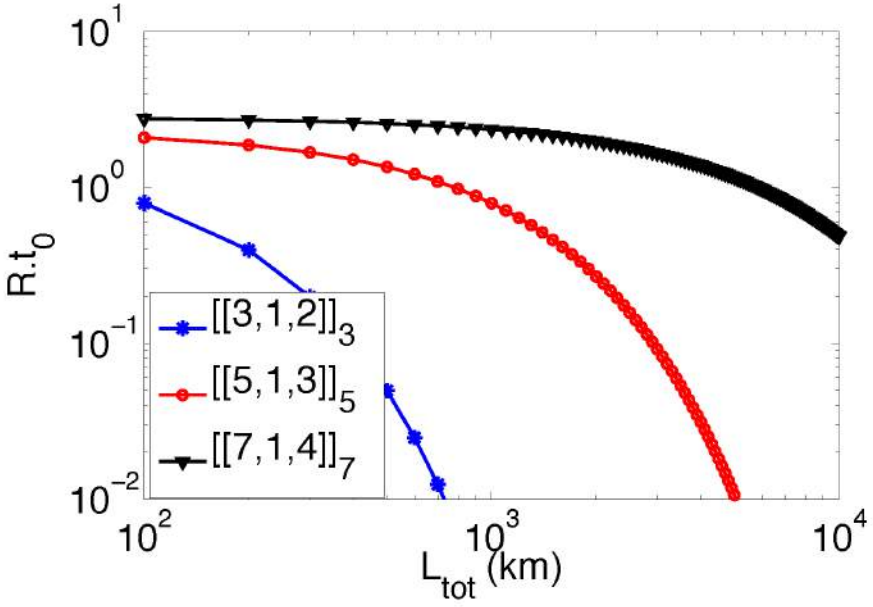
Muralidharan, Zou, Li, Wen, L. J., arXiv:1504.08054

Quantum Polynomial Code

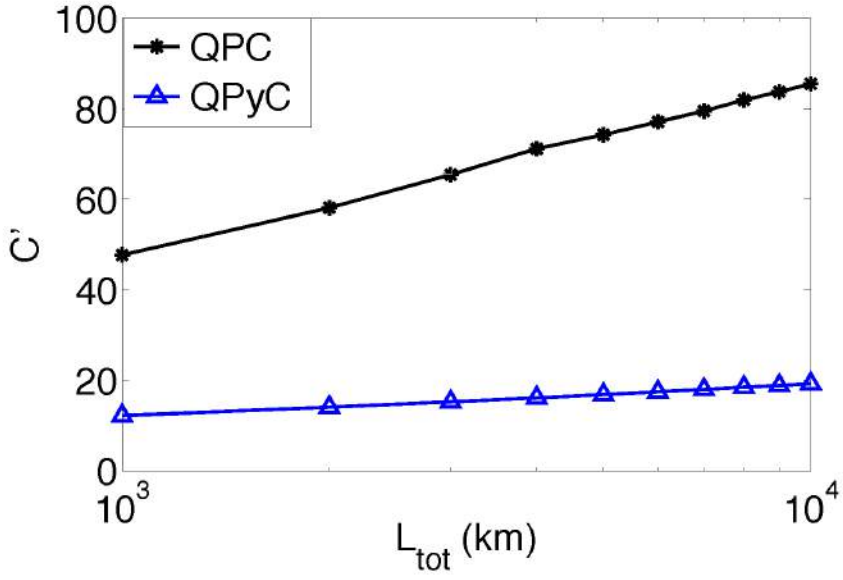
- **Key Idea:** Using d-level quantum system (qudit) for encoding
- **Application of QPyC** for 3rd Gen of QRs:
  - Very good code to correct loss errors
  - More efficient than the PQC code



Efficient QPyC correcting Loss Errors (with 1 km spacing)



Comparison between QPC and QPyC



Muralidharan, Zou, Li, Wen, L. J., arXiv:1504.08054

Continuous Variable Codes

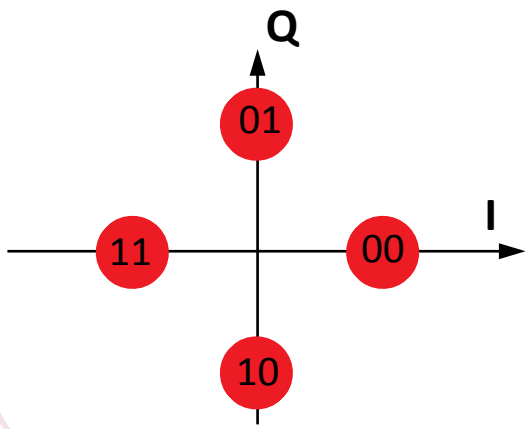
Classical Phase-Shift Keying (PSK):

- Convey data by modulating phase of a pulse signal

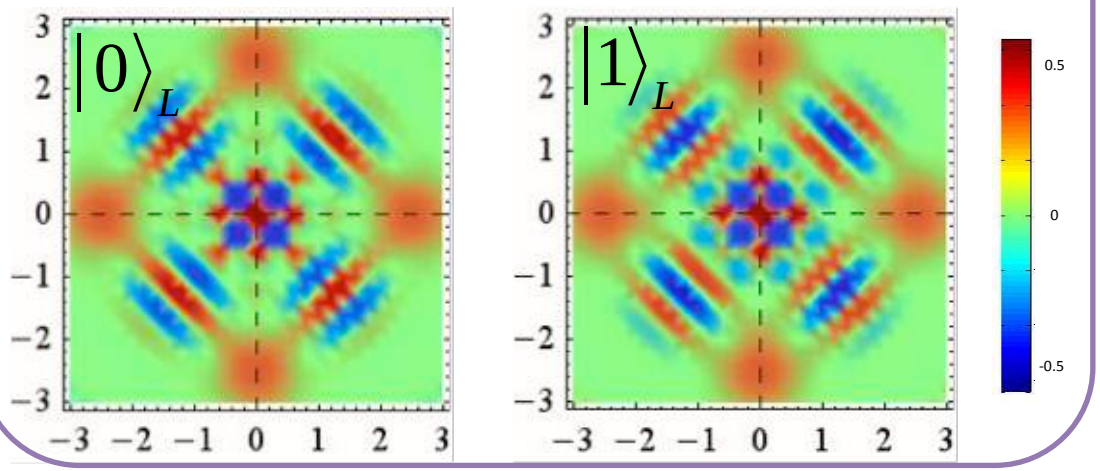
Quantum Cat Codes:

- Superposition of coherent states
- Fixed parity logical subspace

Classical 4-PSK



Quantum Cat Codes



Mirrahimi, Leghtas, Albert, Touzard, Schoelkopf, L. J., Devoret, New J. Phys. **16**, 045014 (2014).

"Cycle of Life" of Cat Codes with Fixed Parity

$$|0_L\rangle = N(|\alpha\rangle + |-\alpha\rangle)$$
$$|1_L\rangle = N(|i\alpha\rangle + |-i\alpha\rangle)$$

No photon loss: $|\Psi^{[0]}\rangle = c_g(|\alpha\rangle + |-\alpha\rangle) + c_e(|i\alpha\rangle + |-i\alpha\rangle)$ **even**

1 photon loss: $|\Psi^{[1]}\rangle = c_g(|\alpha\rangle - |-\alpha\rangle) + ic_e(|i\alpha\rangle - |-i\alpha\rangle)$ **odd**

2 photon loss: $|\Psi^{[2]}\rangle = c_g(|\alpha\rangle + |-\alpha\rangle) - c_e(|i\alpha\rangle + |-i\alpha\rangle)$ **even**

3 photon loss: $|\Psi^{[3]}\rangle = c_g(|\alpha\rangle - |-\alpha\rangle) - ic_e(|i\alpha\rangle - |-i\alpha\rangle)$ **odd**

4 photon loss: $|\Psi^{[0]}\rangle = c_g(|\alpha\rangle + |-\alpha\rangle) + c_e(|i\alpha\rangle + |-i\alpha\rangle)$ **even**

Can overcome:
- **Single photon Loss error**

Properties:

1. Logical states have even photon numbers.
   - **Detect** single photon loss by measuring parity
2. Coherence still maintained after single photon loss
   - **Only need to track** single photon losses

Mirrahimi, Leghtas, Albert, Touzard, Schoelkopf, L. J., Devoret, New J. Phys. **16**, 045014 (2014).

Novel Applications: Quantum Network of Clocks

Quantum Network of Clocks

Question: How to use all the atomic clocks mostly efficiently?

Key Idea: Create nonlocal GHZ state

$$|\Psi_{init}\rangle = |0\rangle^{\otimes N} + |1\rangle^{\otimes N}$$

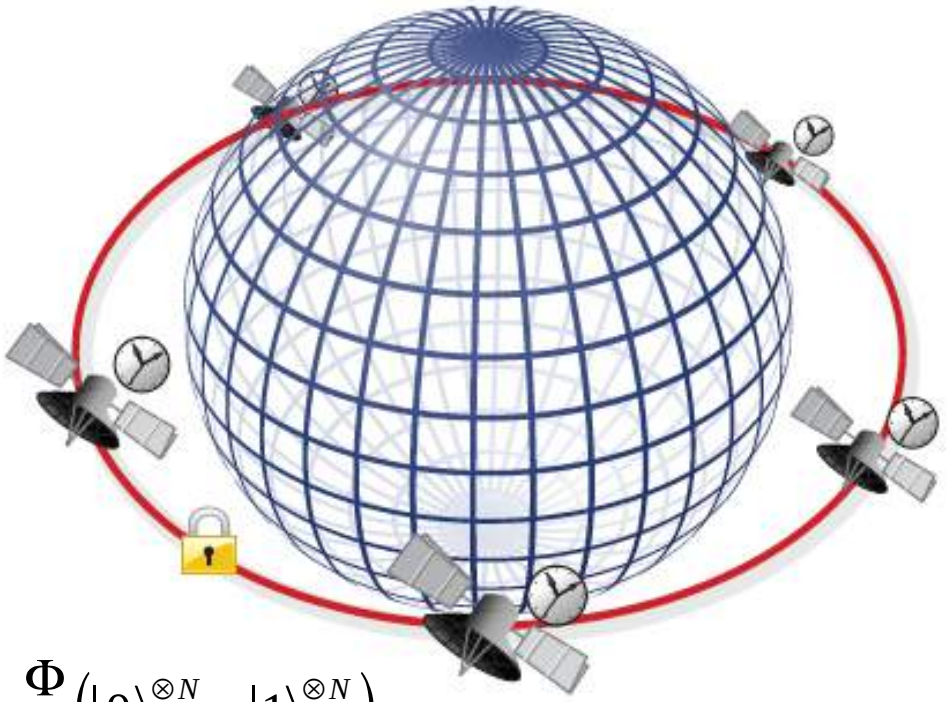
Interrogation

$$|\Psi_{final}\rangle = |0\rangle^{\otimes N} + e^{i\Phi}|1\rangle^{\otimes N}$$

$$\text{with } \Phi = N(\nu_{COM} - \omega_0)T$$

Local Measurement

$$|\Psi_{final}\rangle = \cos\frac{\Phi}{2}\left(|0\rangle^{\otimes N} + |1\rangle^{\otimes N}\right) - i\sin\frac{\Phi}{2}\left(|0\rangle^{\otimes N} - |1\rangle^{\otimes N}\right)$$



$\left(|+\rangle+|-\rangle\right)^{\otimes N} + \left(|+\rangle-|-\rangle\right)^{\otimes N}$ has even # of $|-\rangle$ outcomes.

$\left(|+\rangle+|-\rangle\right)^{\otimes N} - \left(|+\rangle-|-\rangle\right)^{\otimes N}$ has odd # of $|-\rangle$ outcomes.

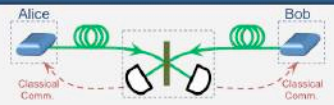
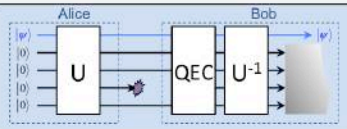
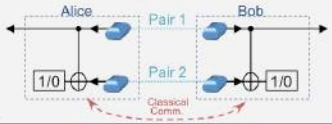
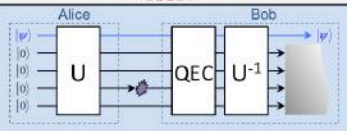
- Optimal use of resources
- Security against sabotage & eavesdropping.

With **local** sampling, one can reliably estimate the value of Φ, and consequently $\nu_{COM} - \omega_0$ with precision ~$1/N$ (v.s. ~$1/\sqrt{N}$ classically).

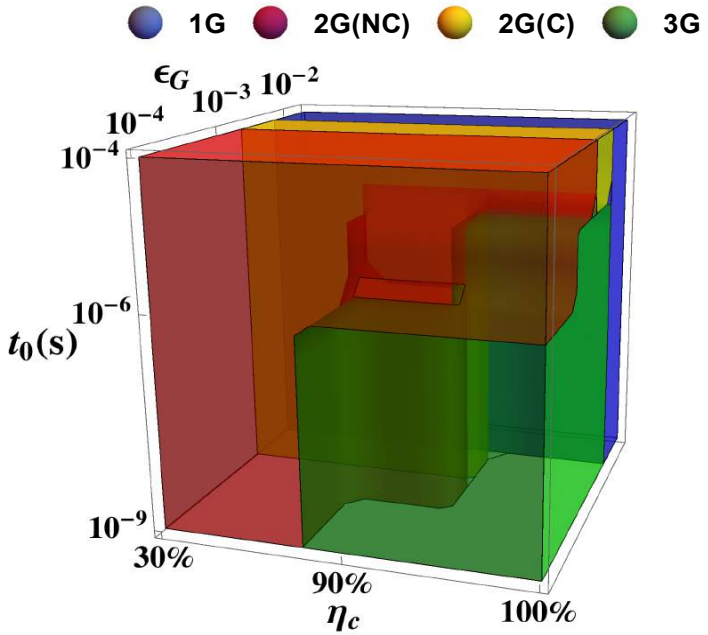
Kómár, Kessler, Bishof, L. J., Sørensen, Ye, Lukin, Nat. Phys. ***10****, 582 (2014); PRL* ***112****, 190403 (2014).*

Summary

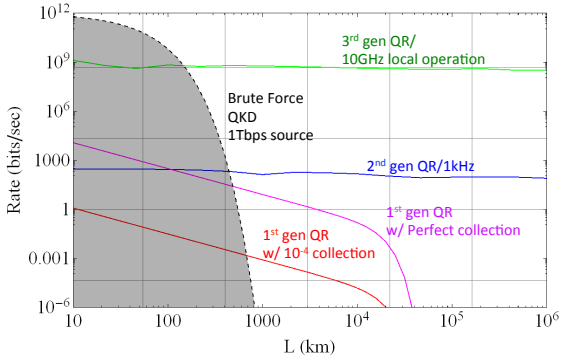
Three Generations of QRs

	Approaches	Example	1st Gen	2nd Gen	3rd Gen
Loss Errors	Heralded Generation [*Two-Way* Comm]		✓	✓	
	Quantum Error Correction [*One-Way* Comm]				✓
Operation Errors	Heralded Purification [*Two-Way* Comm]		✓		
	Quantum Error Correction [*One-Way* Comm]			✓	✓

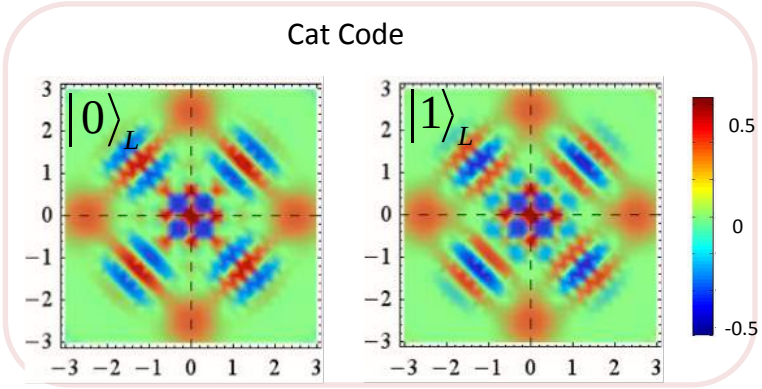
Compare Cost Coefficients



Compare Key Generation Rate



Quantum Codes for QRs



Quantum Network of Clocks

